# Experimental Evaluation of an Exact Algorithm for the Orthogonal Art Gallery Problem

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**Abstract.** We consider the Orthogonal Art Gallery problem (OAGP) whose goal is to minimize the number of vertex guards required to watch an art gallery whose boundary is an $n$-vertex orthogonal polygon $P$. Here, we explore an exact algorithm for OAGP, which we proposed in [1], that iteratively computes optimal solutions to Set Cover problems (SCPs) corresponding to discretizations of $P$. While it is known [1] that this procedure converges to an exact solution of the original continuous problem, the number of iterations executed is highly dependent on the way we discretize $P$. Although the best theoretical bound for convergence is $\Theta(n^3)$ iterations, we show that, in practice, it is achieved after only a few of them, even for random polygons of hundreds of vertices. As each iteration involves the solution of an SCP, the strategy for discretizing $P$ is of paramount importance. In this paper, we carry out an extensive empirical investigation with five alternative discretization strategies to implement the algorithm. A broad range of polygon classes is tested. As a result, we are able to significantly improve the performance of the algorithm, while maintaining low execution times, to the point that we achieve a fivefold increase in polygon size, compared to the literature.

## 1 Introduction

The classical *Art Gallery Problem* originally posed by Victor Klee in 1973 consists in determining the minimum number of guards sufficient to cover the interior of an $n$-wall art gallery [2]. Chvátal showed, in what became known as *Chvátal's Art Gallery Theorem*, that $\lfloor n/3 \rfloor$ guards are occasionally necessary and always sufficient to cover a simple polygon with $n$ vertices [3].

Many variants of the art gallery problem have been studied in the literature. In this paper, we study the variation of the classical art gallery problem that deals specifically with orthogonal polygons (edges parallel to the $x$ or $y$ axis) where guards can only be placed on vertices that define the outer boundary of the gallery. This is called the *Orthogonal Art Gallery Problem* (OAGP) and

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is an important subclass, due to most real life buildings and galleries being orthogonally shaped [4].

The earliest major result concerning this problem, due to Kahn *et al.* [5], states that $\lfloor \frac{n}{4} \rfloor$ guards are occasionally necessary and always sufficient to cover an orthogonal polygon with $n$ vertices. Later, Schuchardt and Hecker proved that minimizing the number of guards in this variation is also NP-hard [6], settling a question that remained open for almost a decade [7].

Several placement algorithms have been proposed in the past, such as Edelsbrunner *et al.* [8] and Sack and Toussaint [7], which deal with the problem of efficiently placing exactly $\lfloor n/4 \rfloor$ guards covering a given orthogonal gallery.

On the other hand, in a recently revised manuscript, based on [9], Ghosh presents an $O(n^4)$ time approximation algorithm for simple polygons yielding solutions within a $\log n$ factor of the optimal. Further approximation results include Eidenbenz [10] who designed algorithms for several variations of terrain guarding problems and Amit *et al.* [11] who analyze heuristics with experimental evidence of good performance in covered area and in the number of guards.

Another approach tackled by Erdem and Sclaroff [12] and Tomás *et al.* in [13] consists of modeling the problem as a discrete combinatorial problem and then solving the corresponding optimization problem. The former discretize the interior of the polygon with a fixed grid, yielding an approximation algorithm and the latter gives empirical analysis of an exact method of successive approximations based on dominance of visibility regions.

Finally, in [1], we presented an exact algorithm to optimally solve the OAGP. In this algorithm, we iteratively discretize and model the problem as a classical *Set Cover problem* (SCP). Besides demonstrating the feasibility of this approach, we showed that, in practice, the number of iterations required to solve instances of up to 200 vertices was very small and that the resulting algorithm turned out to be quite efficient.

**Our contribution.** Though the number of iterations executed by the exact algorithm we proposed in [1] was shown to be polynomially bounded, its practical performance is much better depending on how the polygon is discretized. This becomes clearer when we notice that at each iteration an instance of SCP, a NP-hard problem, has to be solved at optimality, in our case, by an Integer Programming (IP) solver.

In this paper, we conduct a thorough experimental investigation concerning the trade-off between the number and nature of discretizing points and the number of iterations, analyzing the practical viability of each approach. Our test data, available in [14], includes multiple instances for each size of the vertex set, for various classes of orthogonal polygons with up to *a thousand* vertices.

The new experimental results significantly surpassed those we reported in [1]. This is due to the exploration of alternative discretization strategies, which allow us to address difficult instances as well as to handle a fivefold increase in the polygon size compared to the literature, while attaining low execution times.

**Organization of the text.** In the next section, we explain the basic ideas that support the algorithm. Section 3 is devoted to the description of the algorithm